THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **Tissues in Action** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(c)** accumulation of food	Phloem blocked.
3	**(b)** glandular epithelium	Secretion.
4	**(b)** sensory neuron, relay neuron and motor neuron	Reflex arc.
5	**(a)** lignin	Lignified walls.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(b)** xylem is intact	Water still moves.
8	**(c)** a single nucleus and spindle shape	Unstriated.
9	**(d)** dendrite to axon	Direction.
10	**(d)** safranin	Red lignin.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Secondary xylem accumulates every year (1); Lignified walls give strength (1).	1+1
12	Xylem upward only, phloem both directions (1); Xylem passive; phloem needs energy (1).	1+1
13	No growth in length or girth (1); No new organs would develop (1).	1+1
14	Place a twig in red ink (1); Xylem gets coloured red (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Potato turns blue-black as it stores starch (1); Carrot gives less blue-black (1); Parenchyma stores food (1).	1+1+1
16	Constant energy demand (1); Continuous contraction (1); Supplies ATP (1).	1+1+1
17	Vascular cambium: secondary xylem and phloem (1); Cork cambium: cork (1); Together form secondary growth (1).	1+1+1
	OR: Dicot stem (1); Vascular bundles in a ring (1); Open bundles with cambium (1).	1+1+1
18	Apical meristem adds height (1); Lateral meristem adds girth (1); Annual rings record growth (1).	1+1+1
	OR: Day: turgid, stoma open (1); Night: flaccid, stoma closed (1); Controls transpiration (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Place a twig in red ink (1); Section shows red xylem (1); Girdle a branch (1); Bulge above the ring (1); Conclusions about xylem and phloem (1).	1+1+1+1+1
	OR: Xerophytes: thick cuticle, sunken stomata (1); Hydrophytes: aerenchyma (1); Mesophytes: normal tissues (1); Root tissues for water (1); Support tissues (1).	1+1+1+1+1
20	Dicot: ring of open bundles (1); Monocot: scattered closed bundles (1); Cambium present in dicots (1); No secondary growth in monocots (1); Examples (1).	1+1+1+1+1
	OR: Plants: meristems regenerate (1); Animals: epithelium regenerates quickly (1); Bone and liver: moderate (1); Neurons: limited (1); Stem cell therapy (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **Tissues in Action** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(b)** vascular cambium towards the inside	Wood.
3	**(c)** fusion of many cells during development	Syncytium.
4	**(c)** synapse	Junction.
5	**(a)** starch in parenchyma	Blue-black.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(b)** outside	Protection.
8	**(d)** cardiac muscle	Contractile.
9	**(d)** damaged nerve cells are not easily replaced	Limited regeneration.
10	**(a)** xylem	Water conduction.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Thick cuticle reduces water loss (1); Sunken stomata reduce transpiration (1).	1+1
12	RBCs live about 120 days (1); WBCs fight infection and die (1).	1+1
13	Sieve tubes lack a nucleus (1); Companion cells control their functions (1).	1+1
14	Bone has a hard matrix and less blood supply (1); Needs new bone cells to deposit minerals (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Skeletal: many nuclei, striated (1); Smooth: single nucleus, unstriated (1); Cardiac: single or two nuclei, striated, branched (1).	1+1+1
16	Chlorenchyma (parenchyma with chloroplasts) (1); Photosynthesis (1); Found in leaves and green stems (1).	1+1+1
17	Jute and flax from the stem sclerenchyma (1); Cotton from the seed hairs (1); Used for fabrics (1).	1+1+1
	OR: Sugary sap in the phloem lies in the bark (1); Xylem water is deeper (1); The cut opens the phloem (1).	1+1+1
18	Stratified squamous epithelium protects (1); Nerve endings sense (1); Sweat glands and blood vessels regulate temperature (1).	1+1+1
	OR: Provides strength (1); Lignin makes it hard (1); Timber is mostly secondary xylem and fibres (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Cardiac muscle: pumping (1); Epithelium lining the chambers (1); Connective tissue: valves and covering (1); Nervous tissue regulates rate (1); Part of the circulatory system (1).	1+1+1+1+1
	OR: Teased or thin section (1); Stain with methylene blue (1); Striated: bands and many nuclei (1); Smooth: no bands (1); Cardiac: branched with discs (1).	1+1+1+1+1
20	Sensory: receptor to spinal cord (1); Relay in spinal cord (1); Motor: to muscle (1); Muscle contracts (1); Reflex is fast and automatic (1).	1+1+1+1+1
	OR: Select explant (1); Sterile nutrient medium with hormones (1); Callus forms (1); Plantlets grown and transferred (1); Applications: orchids, bananas, conservation (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **Tissues in Action** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** A is true but R is false.	A is true; R is false: cardiac muscles are involuntary.
2	**(b)** its age	One ring per year.
3	**(b)** transparent epithelium and connective tissue	Transparent.
4	**(d)** electrical and chemical changes	Impulse.
5	**(d)** xylem carries water	Transport.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(c)** favourable growth conditions	Good season.
8	**(d)** sweep dust and mucus	Cleaning.
9	**(c)** cardiac muscle	Heart.
10	**(c)** wax-like cutin	Waterproof.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Oxygen and glucose for energy (1); Remove carbon dioxide and wastes (1).	1+1
12	Muscles need much energy for contraction (1); Mitochondria release the energy (1).	1+1
13	Nervous: conduction of impulses (1); Muscular: contraction and movement (1).	1+1
14	Reduces friction (1); Absorbs shock (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Epidermis admits light (1); Palisade tissue photosynthesises (1); Xylem brings water, phloem takes away sugars (1).	1+1+1
16	Immune system recognises foreign tissue (1); Rejection can occur (1); Matching reduces rejection (1).	1+1+1
17	Nervous tissue (1); Impulses cannot pass to muscles (1); Paralysis and loss of sensation (1).	1+1+1
	OR: Biceps contracts, triceps relaxes (1); Tendons pull the forearm bone (1); Hinge joint with ligaments and cartilage (1).	1+1+1
18	Can differentiate into many cell types (1); Replace damaged cells (1); Used in bone marrow transplant (1).	1+1+1
	OR: Cell: gland cell (1); Tissue: epithelium (1); Organ: stomach; system: digestive (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Hinge, ball and socket, pivot (1); Cartilage, synovial fluid, ligaments (1); Muscles and tendons (1); Arthritis: inflammation (1); Wear of cartilage (1).	1+1+1+1+1
	OR: Cells specialise to form tissues (1); Tissues form organs (1); Organs form systems (1); Plants: xylem, phloem, meristems (1); Animals: epithelial, muscle, nerve, connective (1).	1+1+1+1+1
20	Root: epidermis with root hairs (1); Stem: epidermis with cuticle (1); Vascular bundles: alternate vs same radii (1); Pith small in roots, large in stems (1); Functions (1).	1+1+1+1+1
	OR: Skin: stratified squamous (1); Intestine: columnar with microvilli (1); Lung: thin squamous (1); Blood vessels: squamous endothelium (1); Structure fits function (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.